

The Employment and Output Effects of Short-Time Work in Germany*

Russell Cooper[†] and Moritz Meyer[‡] and Immo Schott[§]

September 8, 2026

We study the employment and output effects of short-time work in Germany. Short-time work is a policy instrument used to prevent layoffs in reaction to adverse shocks. It encourages firms to reduce labor demand via the intensive margin (hours worked per employee) instead of the extensive margin (laying off workers). Beginning with the 2009/10 recession, this policy was significantly extended to mitigate the consequences of the financial crisis and later Covid. Using German micro-level data, we estimate a search model with heterogeneous multi-worker firms. Our findings suggest that while short-time work dampens the recessionary rise in unemployment, it deepens the output contraction. In addition to the direct effect of lower hours, the policy slows the reallocation of workers toward more productive firms.

*We thank Michael Burda, Thomas Lemieux, Baris Kaymak, Ryan Michaels, Guido Menzio, Ben Pugsley, and participants at various seminars and conferences for helpful comments. This research project has received funding from the European Union's 7th Framework Programme (FP7/2007-2013) under grant agreement 262608, DwB - Data without Boundaries. We declare that we have no relevant or material financial interests that relate to the research described in this paper. The findings, interpretations, and conclusions expressed in this work do not necessarily reflect the views of the Federal Reserve Board or the World Bank or any affiliated organizations, its Board of Executive Directors, or the governments they represent. The World Bank does not guarantee the accuracy of the data included in this work.

[†]China Economic Research Institute, Liaoning University, European University Institute, and University of Austin. russellcoop@gmail.com

[‡]The World Bank, moritz.meyer@gmail.com

[§]Federal Reserve Board. Division of International Finance, 20th and C Streets, NW, Washington, DC 20551, United States. E-mail: immoschott@gmail.com

1. Introduction

In 2009 Germany experienced one of the largest GDP declines among OECD countries. The left panel of Figure 1 shows that in the first quarter of 2009 Germany's GDP growth was -7.0% (OECD average: -4.7%). Yet there was only a very small uptick of the German unemployment rate during the downturn, as the middle panel of Figure 1 shows. This stands in sharp contrast to other OECD economies, e.g., the United States, shown as the dashed red line in Figure 1, where unemployment rates increased markedly after 2008. At the same time, German labor productivity sharply contracted, whereas U.S. labor productivity increased, as the right panel of Figure 1 shows.

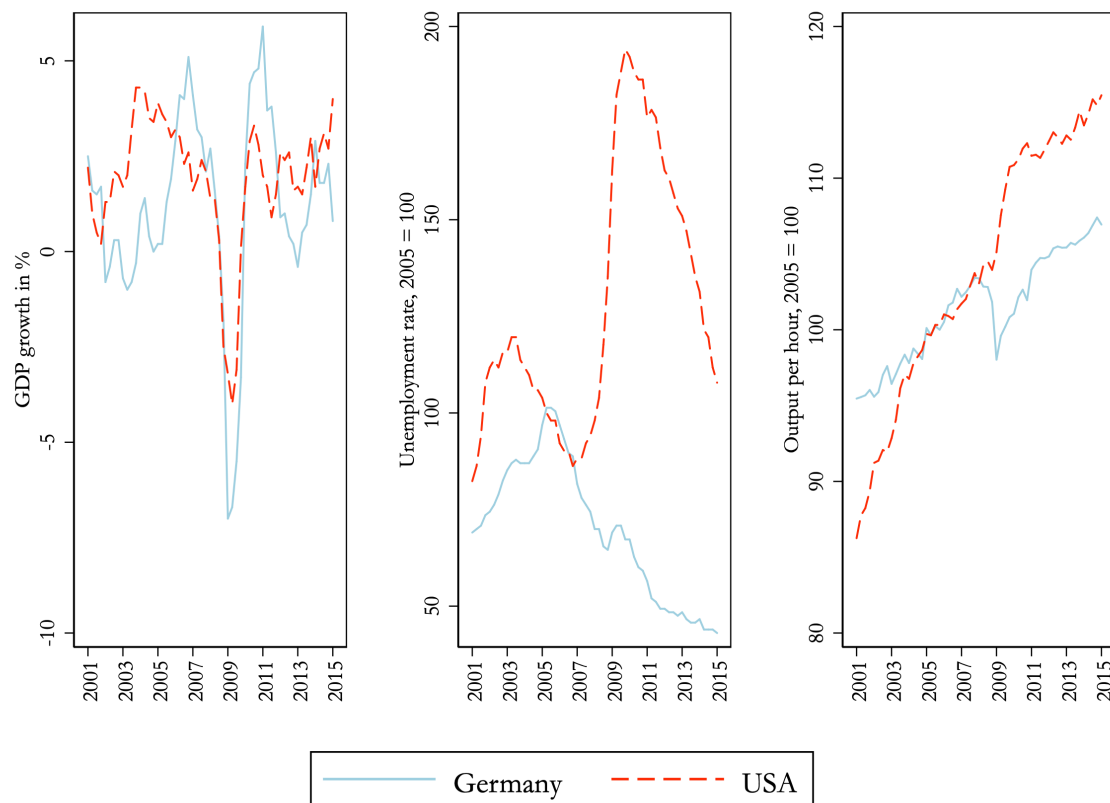


Figure 1: GDP growth and unemployment rates

Note.— GDP growth rates are in % compared to the same period one year ago. Unemployment rates and labor productivity are normalized to 100 in 2005. *Source:* OECD, BLS, German Statistical Office.

The mild increase of the German unemployment rate by less than one percentage point despite the sizeable reduction in GDP growth during 2009/10 has been referred to as the 'German miracle'. This 'miracle' has often been attributed to the use of *Kurzarbeit* (short-time work), which is a policy whose stated primary purpose is to preserve jobs during severe economic downturns. The policy was heavily used during

2009 and during the Covid-induced recession in 2020.¹ In this paper we study the macroeconomic consequences of short-time work in a dynamic general equilibrium model. We find that while short-time work mitigates the impact of recessions on unemployment, it can lock workers into relatively unproductive firms, prevent labor reallocation, and thereby generate negative aggregate productivity effects.

Kurzarbeit is a policy instrument that incentivizes firms to adjust labor demand by reducing employees' working hours rather than resorting to layoffs. Firms that use short-time work reduce hours worked per employee and pay workers in proportion to the actual hours worked. Workers on short-time work are eligible for government compensation of their net earnings loss due to the reduction in hours worked. Workers benefit from this policy because it avoids unemployment. Firms benefit because matches with existing employees are preserved, thereby lowering future hiring and training costs. In 2009/10 and again in 2020, the *Kurzarbeit* program was significantly expanded. Eligibility criteria were loosened, the maximum duration was lengthened from six months to two years, and the availability of the program was extended to part-time workers and apprentices.²

Figure 2 shows the number of workers on short-time work over time. At the height of the 2009/10 recession, approximately 1.5 million workers were on short-time work. More firms used short-time work in 2009 than at any previous point up to that date, including the German reunification. During 2009 the policy's use was heavily concentrated in the manufacturing sector. During 2020, the number of workers using the policy reached new highs, with a total of six million workers on *Kurzarbeit*. Absent large crises, short-time work played a very minor role, e.g., fewer than 25,000 workers were on short-time work during 2008.

To study the consequences of short-time work, we develop a dynamic equilibrium model with heterogeneous multi-worker firms that are subject to search frictions and persistent aggregate and idiosyncratic productivity shocks. Firms choose the number of workers and average hours worked. A key object of the model is the compensation function which is negotiated between a firm and its workers. When available, short-time work incentivizes firms to reduce labor demand through reductions in average hours rather than the number of workers. While this mitigates the impact of recessions on unemployment, it can lock workers into relatively unproductive firms, prevent labor reallocation, and thereby generate negative aggregate productivity effects. We calibrate the model to the German economy prior to the 2009 recession using plant-level data on the universe of German manufacturing plants. The model is then used for a series of quantitative experiments in which we simulate counterfactual economies. We find that, after one year, the unemployment rate is about 0.9 percentage point lower when short-time work is available. At the same time, short-time work deepens the trough in aggregate output from 2.0% to 4.1%, reflecting both the direct reduction in hours and the policy's adverse effects on labor reallocation.

¹We discuss the Covid recession in further detail below.

²For more details about the German institutional background, see Appendix A. An overview for other countries is provided in Giupponi and Landais (2022).

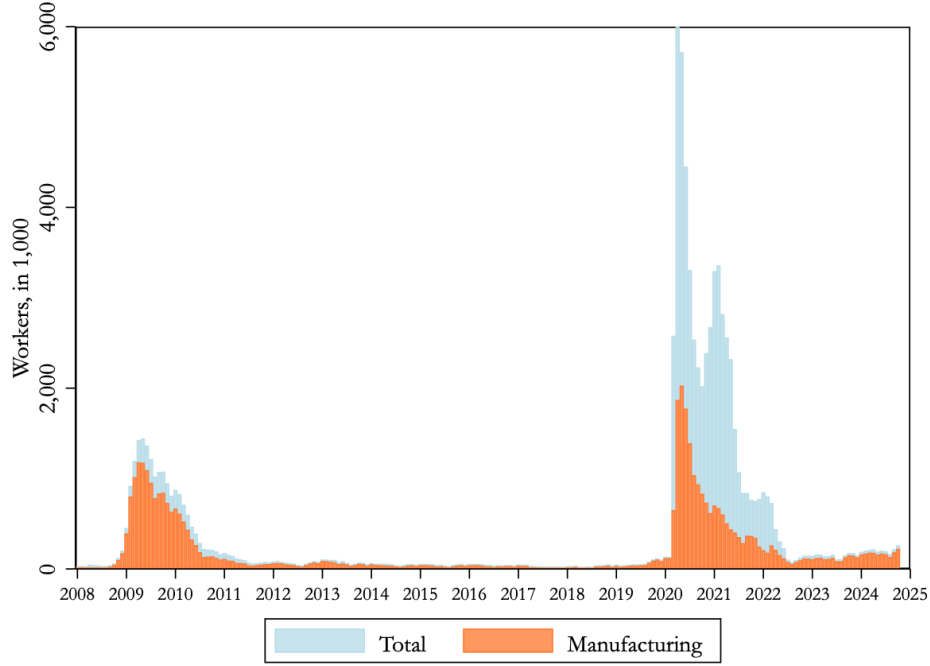


Figure 2: Workers on short-time work

Note.— The total number of workers on short-time work is shown in blue, those in the manufacturing sector are shown in red. At the height of the program in 2009, almost 87% of all short-time workers were employed in manufacturing. *Source:* German Employment Agency.

Related literature A large empirical literature has shown that short-time work schemes can mitigate the impact of the recession on unemployment (Cahuc and Carcillo, 2011; Boeri, Bruecker, Fuchs-Schündelin, and Mayer, 2011; Cahuc, Kramarz, and Nevoux, 2018; Gehrke and Hochmuth, 2020; Kopp and Siegenthaler, 2021). Giupponi and Landais (2022) exploit quasi-experimental eligibility variation in Italy and find that greater local short-time work exposure lowers employment growth and worker inflows at untreated firms and reduces local TFP growth. More broadly, the efficiency with which factors are allocated across production sites plays an important role in aggregate productivity (Hsieh and Klenow, 2009; Bartelsman, Haltiwanger, and Scarpetta, 2013; Cooper and Schott, 2023). By preventing labor from flowing toward more productive firms, short-time work can adversely affect aggregate output through a ‘reallocation channel’. Capturing this channel requires a model that explicitly accounts for heterogeneity in firm size and productivity.

Our main contribution lies in developing such a quantitative dynamic, structural, equilibrium model with heterogeneous firms, which we use to assess the aggregate output, employment, and productivity effects of short-time work. Our model builds on previous work by Cooper, Haltiwanger, and Willis (2007) and Elsby and Michaels (2013), who study heterogeneous multi-worker firms in a search model. As in Elsby and Michaels (2013), we include a rich bargaining framework into a search model with heterogeneous

firms with persistent firm-specific profitability shocks. Relative to Elsby and Michaels (2013), Cosar, Guner, and Tybout (2016), Fujita and Nakajima (2016), and Roys (2016), we add an hours dimension to firms' labor demand, following Cooper et al. (2007).³ Balleer, Gehrke, Lechthaler, and Merkl (2016) is a related paper that also develops a model to study Germany's 2009 *Kurzarbeit* program. Different from their paper, we model heterogeneous firms to allow for a reallocation channel through which labor can be locked into unproductive firms as a consequence of short-time work. Furthermore, we allow firms to use both the intensive and the extensive margins of labor adjustments, even absent short-time work. The elasticity of both margins of labor demand to firm-level shocks is important for determining the effects of the policy.

The rest of the paper is organized as follows. Section 2 describes the model. Section 3 describes the calibration of the model and presents our quantitative results.

2. Model

Our model features multi-worker firms that compete for labor in a frictional labor market. Firms are heterogeneous in terms of productivity and the number of workers they employ. Unemployed workers and vacancies meet according to a constant-returns-to-scale matching technology $M = m(U, V)$, where U is the number of unemployed workers and V is the number of vacancies posted by firms. Firms that post vacancies are matched to a worker with probability $q = M/V$. Unemployed workers find new jobs with probability $\phi = M/U$. A sufficient statistic for the vacancy-filling and job-finding probabilities is labor-market tightness $\theta \equiv V/U$. Worker compensation is the outcome of Nash bargaining between a firm and its workers.

There are two components of labor demand: an intensive margin and an extensive margin. On the extensive margin, firms must decide how many workers to employ. Adjusting the number of workers, either by posting vacancies or by firing, takes one period, which we take to be a month. On the intensive margin, the firm decides average hours worked for its existing workers. Because hours adjustments occur instantaneously, this margin allows firms to react to contemporaneous changes in profitability.

For analytical and notational tractability, we first describe the model without short-time work, derive an analytical solution of the compensation function, and characterize the trade-offs that determine a firm's choice of hours and employment. In Section 2.5 we introduce short-time work to the model.

2.1. Workers

Workers are risk-neutral, with preferences defined over consumption and leisure. Disutility from working h hours is given by $e(h)$, with $e(0) = 0$, $e'(h) > 0$ and $e''(h) > 0$. We

³Other related papers include Burdett and Wright (1989), Smith (1999), Kudoh and Sasaki (2011), and Acemoglu and Hawkins (2014). The models in these papers are static, without idiosyncratic or aggregate profitability shocks. Kudoh, Miyamoto, and Sasaki (2019) study the hours dimension of labor demand in a multi-worker firm model without endogenous job destruction.

assume the functional form $e(h) = e_0 \frac{h^{1+e_1}}{1+e_1}$, where $e_0 > 0$ is a constant and $e_1 > 0$ is the inverse Frisch elasticity. Workers discount the future at rate β .

At the beginning of a period, a worker can be employed or unemployed. The value of being employed in a firm in state $s = \{\varepsilon, n\}$, that is, with productivity ε and n workers, and given an aggregate state S , defined below, is given by

$$W^e(s; S) = \omega(s, h; S) - e(h) + \beta \mathbb{E}_{\varepsilon', S' | \varepsilon, S} [s W^u(S') + (1-s) W^e(s'; S')], \quad (1)$$

which includes the compensation $\omega(s, h; S)$ for working h hours. Hours worked are chosen by the firm. A match can be (endogenously) severed with probability s . In that case, a worker becomes unemployed next period and receives the value of being unemployed, W^u . With the counter-probability $1-s$ the worker remains employed in the firm, which will then be in idiosyncratic state $s' = \{\varepsilon', n'\}$.

The value of being unemployed is given by

$$W^u(S) = b + \beta \mathbb{E}_{S' | S} [(1-\phi) W^u(S') + \phi \mathbb{E}_{\varepsilon', n' | S'} W^e(s'; S')]. \quad (2)$$

The instantaneous utility for an unemployed worker is given by an (exogenous) flow value b . With probability $1-\phi$ a worker does not find a job in the current period and remains unemployed next period. With the counter-probability a match is formed with a currently hiring firm. That firm will be in idiosyncratic state s' next period. The outer expectation in (2) is over future aggregate states, while the inner expectation is over the distribution of (ε', n') among hiring firms in that aggregate state.

2.2. Firms

A unit mass of risk-neutral firms uses labor to produce a homogeneous good. Total labor input is the product of the number of workers, n , and average hours, h . A firm's revenue function $F(n \times h)$ is assumed to be strictly increasing and strictly concave. For the remainder of the analysis we assume $F(n \times h) = (nh)^\alpha$ with $0 < \alpha < 1$. Firms are subject to persistent, time-varying idiosyncratic productivity shocks ε that evolve according to $\log \varepsilon' = (1-\rho_\varepsilon)\mu_\varepsilon + \rho_\varepsilon \log \varepsilon + \sigma_\varepsilon \eta$, where $\eta \sim N(0, 1)$ and μ_ε is the unconditional mean of $\log$ productivity. Let $G(\varepsilon' | \varepsilon)$ denote the conditional distribution of firm productivity.

At the beginning of a period, a firm's individual state s includes the current level of idiosyncratic productivity ε and the current number of employees n , the result of past hiring and firing decisions. Firms choose average hours worked h , taking as given the state-contingent compensation function $\omega(s, h; S)$, derived below. Production occurs, with labor input determined by the existing n workers and the chosen average hours h . At the end of the period, before future productivity is known, the firm chooses future employment n' . Firms discount the future at the same rate as households, β . The aggregate state of the economy is given by S and includes aggregate productivity A and labor market tightness θ . The evolution of the aggregate state will be explained in more detail below.

Formally, the problem of a firm can be written recursively as follows:

$$V(s; S) = \max_{h, n'} A\varepsilon (nh)^\alpha - \omega(s, h; S) \cdot n - C(n', n; S) + \beta \mathbb{E}_{\varepsilon', S' | \varepsilon, S} V(s'; S'), \quad (3)$$

where the function $C(n', n; S)$ represents the costs of adjusting the number of workers. To hire new workers, firms post vacancies at a linear cost c_v . Because each vacancy is filled with probability $q(S)$, a firm wanting to add $n' - n$ expected hires needs to post $(n' - n)/q(S)$ vacancies. The vacancy-filling probability is an equilibrium object taken as given by firms. By firing f workers, next period's employment becomes $n' = n - f$. There are no firing costs and no exogenous quits.⁴ Labor adjustment costs take the form

$$C(n', n; S) = \frac{n' - n}{q} \cdot c_v \cdot \mathbb{1}^v, \quad (4)$$

where $\mathbb{1}^v$ is an indicator function that takes the value of one if a firm is posting vacancies, i.e., when $n' > n$.

We now derive the optimal hours and employment choices given the compensation function. Then the compensation function is determined from the outcome of the bargaining process.

Hours Hours are chosen contemporaneously to maximize current period profits. From (3) the optimality condition for average hours worked is

$$A\varepsilon \alpha h^{\alpha-1} n^\alpha = \omega_h(s, h; S) \cdot n. \quad (5)$$

The firm chooses hours to equalize the marginal benefit from increasing total labor input by an additional hour with the marginal cost, which is that all n workers must be compensated for one additional hour of work. The derivative of the compensation function with respect to changes in average hours is denoted as ω_h .

Employment From (3) the first order condition for future employment holds when the firm is actively adjusting employment, i.e., $n' \neq n$. For firms posting vacancies

$$\beta \mathbb{E}_{\varepsilon', S' | \varepsilon, S} V_n(\varepsilon', n'; S') = \frac{c_v}{q(S)} \quad \text{if } n' > n \quad (6)$$

and for firms that are firing

$$\beta \mathbb{E}_{\varepsilon', S' | \varepsilon, S} V_n(\varepsilon', n'; S') = 0 \quad \text{if } n' < n, \quad (7)$$

where $V_n(\varepsilon', n'; S')$ denotes the partial derivative of the future value function with respect to future employment. Because optimal employment is increasing in productivity, these conditions imply two productivity cutoffs for each (n, S) . The lower cutoff $\underline{\varepsilon}(n; S)$ is the

⁴Linear hiring and firing costs both create regions of inactivity in employment adjustment. Firing costs therefore have qualitative and quantitative implications that are very similar to the model presented here.

productivity level at which the firm is indifferent between leaving employment unchanged and reducing it. Firms below $\underline{\varepsilon}(n; S)$ fire workers until (7) holds with equality. The upper cutoff $\bar{\varepsilon}(n; S)$ is the productivity level at which the firm is indifferent between leaving employment unchanged and increasing it. This implies that firms above $\bar{\varepsilon}(n; S)$ post vacancies until (6) holds with equality. Firms between the two cutoffs leave employment unchanged.

2.3. Compensation

Because of labor market frictions, quasi-rents exist from forming a match. This gives rise to a compensation function which depends on a firm's productivity level and labor demand choices. The compensation function is obtained through a Nash bargain between a firm and its workers. Employment adjustment and wage bargaining occur at the end of the period. After the firm chooses next period's employment n' , the firm and its n' workers bargain over a state-contingent compensation schedule for the following period, before future aggregate and idiosyncratic productivity are realized. Accordingly, the compensation $\omega(s, h; S)$ that enters current profits was negotiated in the preceding period for the inherited workforce n . Once the current state is realized, the firm chooses hours according to (5), taking this state-contingent schedule as given. Hiring costs are sunk by the time bargaining occurs.

Because the production technology features decreasing returns to scale, we adopt the bargaining solution first proposed in Stole and Zwiebel (1996). The key difference to standard Nash bargaining is that hiring an additional worker also affects the marginal product of labor of all other workers. Every worker is therefore treated as marginal in the negotiations.⁵ The bargain takes as given that firms will optimally select hours according to (5).

Bargaining takes place over the surplus generated by the marginal worker. If negotiations fail, the worker becomes unemployed and receives the value $W^u(S)$. The firm loses the marginal worker, reducing its value by $V_n(s; S)$. Thus, the worker's surplus from reaching an agreement is $W^e(s; S) - W^u(S)$, while the firm's surplus from retaining the marginal worker is $V_n(s; S)$. Nash bargaining implies

$$W^e(s; S) - W^u(S) = \frac{\eta}{1 - \eta} V_n(s; S),$$

where η denotes the worker's bargaining power. The compensation function solves the differential equation

$$\omega(s, h; S) = (1 - \eta) [e(h) + b] + \eta [A\varepsilon\alpha h^\alpha n^{\alpha-1} - \omega_n(s, h; S)n + \theta c_v], \quad (8)$$

with solution

$$\omega(s, h; S) = (1 - \eta) [e(h) + b] + \eta \left[\frac{A\varepsilon\alpha h^\alpha n^{\alpha-1}}{1 - \eta(1 - \alpha)} + \theta c_v \right]. \quad (9)$$

⁵This bargaining protocol is commonly used in models with multi-worker firms, i.e., Smith (1999); Elsbey and Michaels (2013); Acemoglu and Hawkins (2014); Roys (2016). The derivation of the compensation function is shown in Appendix C.1.

If workers had no bargaining power, i.e., $\eta = 0$ then the compensation function would make them indifferent between working and unemployment. A positive bargaining power implies that workers obtain a share of the marginal product of labor. Higher vacancy creation costs increase compensation because they make failed negotiations more costly for the firm.

2.4. Optimal choices

Given the expression for the compensation function in (9), we can now further analyze firms' optimal policy functions for labor demand.

Hours The term $\omega_h(\varepsilon, n, h)$ in the first-order condition for hours, (5), is

$$\omega_h(s, h; S) = (1 - \eta) e'(h) + \eta \frac{A\varepsilon\alpha^2 h^{\alpha-1} n^{\alpha-1}}{1 - \eta(1 - \alpha)} > 0. \quad (10)$$

By increasing hours, firms must compensate workers for the higher disutility of work. In addition, with $\eta > 0$, workers share part of the marginal product of labor. Combining this expression with (5) and using the fact that $e'(h) = e_0 h^{e_1}$, we obtain the optimal hours policy

$$h(s; S) = \left(\frac{A\varepsilon\alpha n^{\alpha-1}}{e_0(1 - \eta(1 - \alpha))} \right)^{\frac{1}{1-\alpha+e_1}}, \quad (11)$$

which is increasing in aggregate and idiosyncratic productivity and decreasing in the existing number of workers.

Although hours can be adjusted contemporaneously, their flexibility is limited by decreasing returns to labor and the increasing marginal compensation associated with the convex disutility of labor.⁶ The terms b and θ_{c_v} in (9) act as fixed components of compensation per worker. These components make additional employees relatively costly, while convex work disutility makes longer work schedules increasingly costly. Together, these forces shape how firms divide labor input between employment and hours. Firms therefore use hours to respond immediately to shocks, but persistent changes in labor demand eventually require adjustments in employment.

Employment For the case of a firm that adjusts employment, i.e., for $n' \neq n$, the optimality conditions for the choice of future employment, (6) and (7), can be summarized as

$$\beta \mathbb{E}_{\varepsilon', S' | \varepsilon, S} V_n(s'; S') = \frac{c_v}{q} \cdot \mathbb{1}^v, \quad (12)$$

where $\mathbb{1}^v$ is the indicator function that takes the value of one when a firm is hiring, and zero otherwise. Using the envelope condition of the value function defined in (3), the

⁶The costs of deviating from 'baseline hours' can be interpreted as capturing constraints associated with working-time accounts or firm-level deviations from collective agreements, see, e.g., (Burda and Hunt, 2011; Dustmann, Fitzenberger, Schönberg, and Spitz-Oener, 2014).

definition of the labor adjustment costs in (4), and combining them with the optimality condition for hours, (11), we obtain

$$V_n(s; S) = A\varepsilon\alpha h^\alpha n^{\alpha-1} - \omega(s, h; S) - \omega_n(s, h; S)n + \frac{c_v}{q} \cdot \mathbb{1}^v, \quad (13)$$

which says that marginally increasing the number of workers today affects the firm's value through an effect on production, the additional compensation payments, a decrease in compensation per worker resulting from a lower marginal product of labor, and hiring costs for firms that are posting vacancies.

Note that it follows from (9) that $\omega_n(s, h; S)$, the marginal effect of employment on compensation in (13), is negative: $\omega_n(s, h; S) = -\frac{(1-\alpha)\eta A\varepsilon\alpha h^\alpha n^{\alpha-2}}{1-\eta(1-\alpha)} < 0$. This gives rise to an *over-employment effect* first analyzed in Smith (1999). Because firms do not take worker compensation as given they can lower the marginal product of labor (and hence the compensation paid to all workers) by increasing employment. In this way the compensation function becomes an important determinant for firms' labor demand.

2.5. Short-time work

We now introduce short-time work to the model. The policy is initially unexpected and can become available only during recessions, defined as $A < 1$. If short-time work is available, only eligible firms can choose to participate. We define *eligible* firms as those that would reduce employment absent the short-time work policy. This criterion captures the spirit of the German Code of Social Law, Book 3 (SGB III), which stipulates that workers on *Kurzarbeit* must have been employed by the same firm prior to the reduction in working time.⁷

Participating firms must keep employment constant but can optimally re-adjust hours per worker. The government subsidizes hours reductions by compensating a part of affected workers' income loss due to reduced hours. The replacement rate is parameterized by $0 \leq \delta < 1$. We now derive worker compensation and the optimal hours policy under short-time work, followed by an eligible firm's participation decision.

Compensation and hours under short-time work We first characterize compensation and hours conditional on participation in the short-time-work program. The compensation schedule is not renegotiated when short-time work is introduced. We denote by $\tilde{\omega}(s, \tilde{h}; S)$ the compensation paid by the firm when hours are reduced to $\tilde{h}$. The government replaces a fraction δ of the resulting earnings loss relative to the no-policy compensation $\omega(s, h; S)$. Worker flow utility under short-time work is given by total worker compensation—the compensation paid by the firm at actual hours worked plus

⁷Eligibility is determined using the employment policy of the economy in which short-time work is unavailable. A firm in state (ε, n) is eligible if its no-policy employment choice, evaluated at the average value of θ during the first year of the no-policy recession, is below its current employment. Holding this value fixed prevents policy-induced movements in tightness from changing firms' eligibility, but is quantitatively unimportant.

the government subsidy—minus the disutility from working $\tilde{h}$ hours:

$$U^{\text{STW}} \equiv \tilde{\omega}(s, \tilde{h}; S) + \delta \cdot \underbrace{\left[\omega(s, h; S) - \tilde{\omega}(s, \tilde{h}; S) \right]}_{\text{income loss due to STW}} - e(\tilde{h}) \quad (14)$$

We assume that compensation under short-time work leaves workers' current-period flow utility at its no-policy level, i.e., the firm captures the subsidy-induced surplus through a reduction in compensation. This assumption can be interpreted as a participating firm making a take-it-or-leave-it offer: the firm bears the cost of applying for the program and sets compensation so that the worker is indifferent, in current-period flow terms, between participation and the no-policy allocation. Let $\bar{U} = \omega(s, h; S) - e(h)$ denote the flow utility received by a worker who works 'normal' hours, i.e., hours absent short-time work. Equalizing $\bar{U}$ with U^{STW} and solving for $\tilde{\omega}(s, \tilde{h}; S)$ results in the compensation paid by firms under short-time work:

$$\tilde{\omega}(s, \tilde{h}; S) = \omega(s, h; S) - \frac{1}{1 - \delta} \cdot \left(e(h) - e(\tilde{h}) \right). \quad (15)$$

It is given by the compensation that would have been paid absent short-time work, minus a term that accounts for the utility gain from working fewer hours. This term is scaled by the generosity δ of the short-time work policy. The more generous the policy, the lower the compensation paid by participating firms, and the larger the fraction of total worker compensation that is paid by the government.

To determine optimal hours under short-time work, we evaluate the optimality condition for hours in (5) at the compensation function under short-time work, (15). This results in the following optimal hours policy under short-time work:

$$\tilde{h}(s; S) = \left(\frac{(1 - \delta)\alpha A \varepsilon n^{\alpha-1}}{e_0} \right)^{\frac{1}{1-\alpha+e_1}}. \quad (16)$$

Compared to 'normal' hours in (11), hours under short-time work are lower because of two factors. First, a higher replacement rate δ induces a larger reduction in hours. Because the government replaces a greater share of the associated earnings loss, the firm can reduce the compensation it pays by more while leaving the worker at the no-policy flow-utility level. Second, the Nash-bargained compensation schedule is not renegotiated under short-time work. Instead, firm-paid compensation is adjusted according to the rule described above. Absent the policy, workers share in the marginal output from longer hours through Nash bargaining, as captured by the term $1 - \eta(1 - \alpha) < 1$ in equation (11). The absence of this bargaining term in (16) lowers hours under short-time work relative to normal hours.

Participation decision We now consider whether an eligible firm chooses to participate in the short-time work program. If the policy is available, eligible firms can pay a fixed, per-period cost κ , scaled by the number of workers, to participate.⁸ A participating

⁸The scaling captures administrative and implementation costs that rise with the number of covered workers and prevents making participation mechanically more attractive for larger firms.

firm's value function is

$$V^{\text{STW}}(s; S) = \max_{\tilde{h}} A\varepsilon \left(n\tilde{h} \right)^{\alpha} - \tilde{\omega}(s, \tilde{h}; S) \cdot n - \kappa n + \beta \mathbb{E}_{\varepsilon', S' | \varepsilon, S} V(s'; S'), \quad (17)$$

where $\tilde{h}$ are hours chosen under short-time work and $\tilde{\omega}(\cdot)$ is per-worker compensation under short-time work. Because participating firms do not adjust employment, there are no employment adjustment costs. The value function of a non-participating firm is

$$V^{\text{no}}(s; S) = \max_{h, n'} A\varepsilon (nh)^{\alpha} - \omega(s, h; S) \cdot n - C(n', n; S) + \beta \mathbb{E}_{\varepsilon', S' | \varepsilon, S} V(s'; S'), \quad (18)$$

with the participation decision determined by

$$V(s; S) = \max \{ V^{\text{STW}}(s; S), V^{\text{no}}(s; S) \}. \quad (19)$$

We denote as $\chi(s, S)$ the indicator function that takes the value of one if a firm is participating in the short-time work program, and zero otherwise.

Employment under short-time work A firm participating in the short-time work program must keep its employment unchanged, so $n' = n$. Even for firms not currently participating in the program, the introduction of short-time work has two effects. First, these firms may participate in the future. This option value makes retaining workers more attractive today. Second, the policy changes the perceived law of motion for labor-market tightness θ . Both effects alter the values and choices of non-participating firms relative to an economy in which short-time work is unavailable.

2.6. Equilibrium

The distribution of firms over idiosyncratic productivity and employment levels is denoted λ . Its law of motion is $\lambda' = \mathbf{H}(\lambda; A, \theta)$, where the mapping $\mathbf{H}$ is determined by the employment policy function $n'(\varepsilon, n; S)$ and the exogenous process for idiosyncratic and aggregate shocks.

The equilibrium consists of policy functions $n'(s; S)$, $h(s; S)$, $\tilde{h}(s; S)$, and $\chi(s; S)$, a negotiated compensation function $\omega(s, h; S)$ and short-time work compensation $\tilde{\omega}(s, \tilde{h}; S)$, an endogenous labor market tightness θ , as well as consistent expectations regarding the future evolution of the aggregate state S , which includes an aggregate productivity level A , the current labor market tightness θ , and the joint distribution λ .

3. Quantitative analysis

This section presents our quantitative results. We fit the model described in Section 2 to German data prior to the large increase in *Kurzarbeit* in 2009, i.e., an economy without the extraordinary expansion of cyclical STW. We use the numerical solution of the model to characterize the tradeoff between hours and employment and how it is affected by short-time work. Our main results show that short-time work preserves employment by shifting labor demand adjustments toward hours, but slows the reallocation of workers toward more productive firms.

3.1. Parameterization

We calibrate a stationary version of the model to the German economy using data from 1995–2008. We use a combination of aggregate time-series moments and plant-level moments from the universe of German manufacturing plants.⁹ All parameter values are summarized in Table 1.

The upper part of Table 1 shows the preassigned parameters. The discount factor is set to $\beta = 0.997$. The parameter α which governs the decreasing returns to scale in production is set to $2/3$.¹⁰ The matching function is assumed to have constant returns to scale, $M = \mu U^\gamma V^{1-\gamma}$. We set the matching elasticity to $\gamma = 0.65$, as estimated by Kohlbrecher, Merkl, and Nordmeier (2016) using German microdata. Worker bargaining power η is set to equal the match elasticity. The average unemployment rate during the 1995–2008 period was 9%. Using monthly data from the German Employment Agency on the number of unemployed and open vacancies, we find an average labor market tightness of $\theta = 0.091$ and a monthly job-finding probability of $\phi = 0.062$. This implies a matching efficiency of $\mu = 0.144$. The resulting average monthly vacancy-filling probability is $q = 0.684$.

We estimate the parameters $\Theta = (\rho_\varepsilon, \sigma_\varepsilon, \mu_\varepsilon, e_1, b, c_v)$ jointly by minimizing the proportional distance between model and data moments. The resulting parameter values are shown in the second panel of Table 1. At each candidate parameter vector, the scale of work disutility, e_0 , is computed from the average-hours normalization. The seven targets, reported in the upper part of Table 2, capture unemployment, the replacement rate, average hours and firm size, the frequency of small adjustments in hours and employment, and the relative cross-sectional dispersion of employment and hours. The moments are chosen to be informative about the underlying parameters, although due to the non-linear nature of the model there is no one-to-one mapping from parameters to moments. The unemployment-rate moment enforces the consistency requirement that the stock and flow unemployment rates be equal.¹¹ We discuss the mapping from parameters to moments in more detail in Appendix C.2.

Inactivity is an important feature of the German data, consistent with the finding by Jung and Kuhn (2014) that worker flows are substantially lower in Germany than in the United States. Table 2 shows that 32% of establishments in the data make annual hours changes smaller than 2.5% in absolute value, compared with 30% in the model; the corresponding shares for employment are 28% and 29%. Thus, the calibrated model closely reproduces the observed prevalence of small adjustments on both margins.

The policy parameters (κ, δ) determine the extensive and intensive margins of short-

⁹The data are described in Appendix B.

¹⁰Our data does not allow us to estimate the parameter α directly from the German plant-level data. Elsby and Michaels (2013) use a value of 0.60, while Cooper et al. (2007) use a value of 0.65. The estimates in Bachmann and Bayer (2014) for a cross-section of German non-financial firms imply a value of $\alpha = 0.7$.

¹¹The unemployment rate pins down equilibrium labor market tightness θ , which must imply $u_{\text{stock}} = u_{\text{flow}}$, where u_{stock} is the unemployment rate implied by the stationary firm distribution and u_{flow} is the unemployment rate implied by the dynamic inflows and outflows from unemployment, $u' = u \cdot (1 - \phi(\theta)) + \xi(\theta)$, i.e., aggregate job finding and job destruction rates.

Table 1: Model Parameters.

Parameter	Meaning	Value	Target/source
<i>Preassigned parameters</i>			
β	Discount factor	0.997	Annual $r = 4\%$
α	Labor elasticity	2/3	Literature
γ	Match elasticity	0.650	Kohlbrecher et al. (2016)
η	Worker bargaining power	0.650	Hosios condition
μ	Matching efficiency	0.144	$\theta = 0.091$
<i>Jointly estimated parameters</i>			
ρ_ε	Persistence of $\log \varepsilon$	0.926	
σ_ε	Innovation std. dev.	0.050	
μ_ε	Mean of $\log \varepsilon$	0.391	
e_1	Disutility of work	1.995	
b	Flow value of unemployment	1.845	
c_v	Vacancy creation cost	0.611	
<i>Internally calibrated parameter</i>			
e_0	Disutility scale parameter	7.2×10^{-5}	Average hours
<i>Policy parameters</i>			
κ	STW fixed cost per worker	0.283	15.4% STW take-up
δ	Generosity of STW program	0.399	28.1% hours reduction

Note.— The parameters in the second panel are estimated jointly for the stationary economy using the moments in the upper part of Table 2. At each candidate parameter vector, e_0 is computed from b , e_1 , and the average-hours normalization. The policy parameters in the lower panel are calibrated separately to the reported targets.

time work (*cf.* equations (19) and (16)). We use two additional data moments from the 2009/10 recession: *i*) the fraction of workers on short-time work during a recession, and *ii*) the average hours reduction of firms using short-time work. Equation (16) permits us to solve analytically for δ from the hours-reduction target. We then choose κ to match worker participation during the first year of a sustained recession. The resulting parameter values are shown in the lower part of Table 1, and the corresponding moments are shown in the lower part of Table 2.

Aggregate productivity can take a number of discrete values whose evolution is governed by a transition matrix Π_A so that on average, booms last for three, and recessions for one year, roughly in line with the German data. Firms need to forecast future labor market tightness, an equilibrium object which depends on the entire firm distribution. Following the literature that builds on Krusell and Smith (1998), we specify the following regression which determines firms' forecast of future labor market tightness:

$$\log \theta' = \beta_0 + \beta_1 \cdot \log \theta + \beta_2 \log A' + \beta_3 (A = A') + \nu \quad (20)$$

Table 2: Data and Model Moments

Moment	Data	Model
Unemployment rate	0.090	0.099
Average hours	34.0	34.6
Average firm size	98.5	99.0
$cv(n)/cv(h)$	5.63	5.89
Small hours changes	0.32	0.30
Small employment changes	0.28	0.29
Replacement rate	0.60	0.63
Workers using STW (in %)	15.4	15.6
Average hours reductions through STW (in %)	28.1	28.1

Note.— Moments used to estimate $\Theta = (\rho_\varepsilon, \sigma_\varepsilon, \mu_\varepsilon, e_1, b, c_v)$ are shown in the upper part. The replacement rate is calculated as the ratio of b to average worker compensation. $cv(n)/cv(h)$ is the ratio of the coefficient of variation of employment and average hours. Small hours and employment changes report the fraction of firms with annual changes of less than 2.5% in absolute value. The lower two rows report the policy moments used to calibrate κ and δ . Sources: German Employment Agency and AFiD-Panel Industriebetriebe.

We iterate numerically over the parameters $\{\beta_0, \beta_1, \beta_2, \beta_3\}$ to find the fixed point. The solution provides a very accurate internal forecast of labor market tightness in the sense that the R^2 of this regression exceeds 0.999 and has a very small maximum forecast error (Den Haan, 2010).¹²

3.2. Optimal hours and employment choices at the firm-level

We now discuss some key properties of the quantitative model. First, we show how the choices of hours and employment without short-time work vary across the firm distribution. Panel (a) of Figure 3 shows that less productive firms choose fewer employees and lower hours. The main reason for the positive correlation between productivity and hours is that labor takes time to build: following a productivity shock, firms initially accommodate changes in labor demand by adjusting hours. Panel (b) shows output per worker y/n and labor productivity $y/(n \cdot h)$. Both measures correlate positively with idiosyncratic firm productivity. The reason lies in the convex disutility from labor, which limits hours adjustments to unexpected productivity changes and implies that changes in labor input do not fully absorb productivity changes.

Figure 4 illustrates how firms choose hours, both with and without short-time work. Evaluated at a given firm state (ε, n) , the solid vertical line indicates the optimal hours choice without short-time work. It is determined by (11) and is represented graphically by the intersection of the green and the light red lines. The green line shows the marginal

¹²We show a sample of the forecasted labor market tightness in Figure C.3 in Appendix C.2.

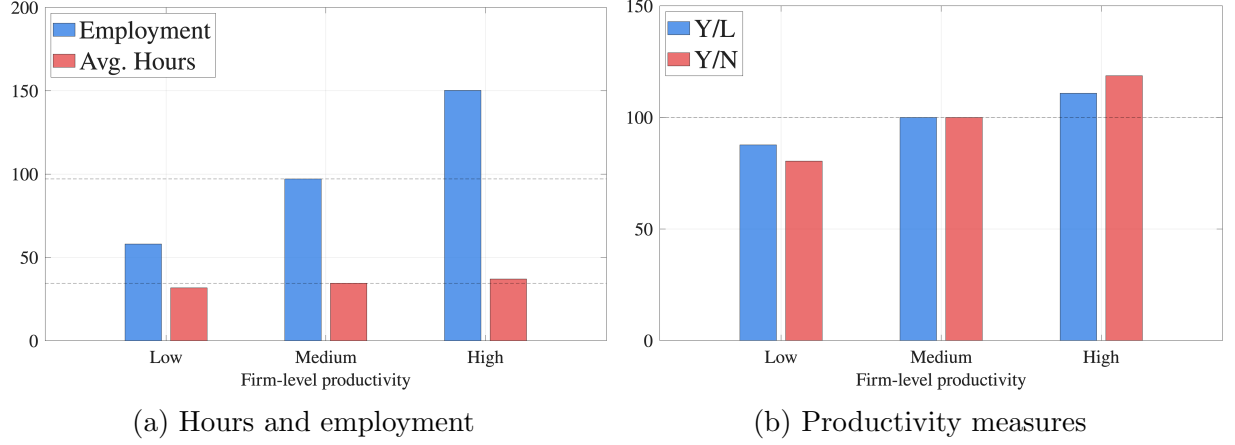


Figure 3: Firm size and productivity

Note.— Firms with idiosyncratic productivity levels equal to the 25th, 50th, and 75th percentile are shown. In panel (a), the horizontal lines are drawn at hours and employment of the median productivity firm. In panel (b), the horizontal line is at the normalized measures for the median productivity firm.

product of working an additional hour and it is downward-sloping because of decreasing returns to scale. The light red line shows the marginal cost of an additional hour, which is increasing over the range of hours shown in the figure, driven by the convex disutility of labor.¹³

Under short-time work, optimal hours are given by (16), indicated by the dotted vertical line. Optimal hours fall because of a change in the marginal-cost schedule under short-time work (as discussed in the text under equation (16)), which is shown by the dark-red line.

3.3. Recessions with and without short-time work

This section presents our main quantitative result. We compare the response to a negative aggregate productivity shock in two economies: one in which short-time work is never available and one in which it is unexpectedly introduced at the onset of a recession. Once introduced, the policy remains available for as long as the recession lasts, and firms correctly anticipate its continued availability conditional on the evolution of the aggregate state.

The two cases are compared in Figure 5. The solid blue lines show the recession without short-time work.¹⁴ Following the negative productivity shock, output falls by about 2% and firms reduce their labor demand. Without short-time work, average hours

¹³We have $\omega_{hh} = (1 - \eta)e_0e_1h^{e_1-1} - \eta \frac{A\varepsilon\alpha^2(1-\alpha)n^{\alpha-1}}{1-\eta(1-\alpha)}h^{\alpha-2}$. The first part is positive because disutility is convex. The second part is negative because the worker's share of the bargaining surplus decreases in hours, due to decreasing returns. By contrast, under short-time work marginal costs must be increasing in hours because $\tilde{\omega}_h = \frac{e'(h)}{1-\delta}$.

¹⁴The size of the negative productivity shock was chosen to be in line with the observed quarterly GDP growth during the first quarter of 2009.

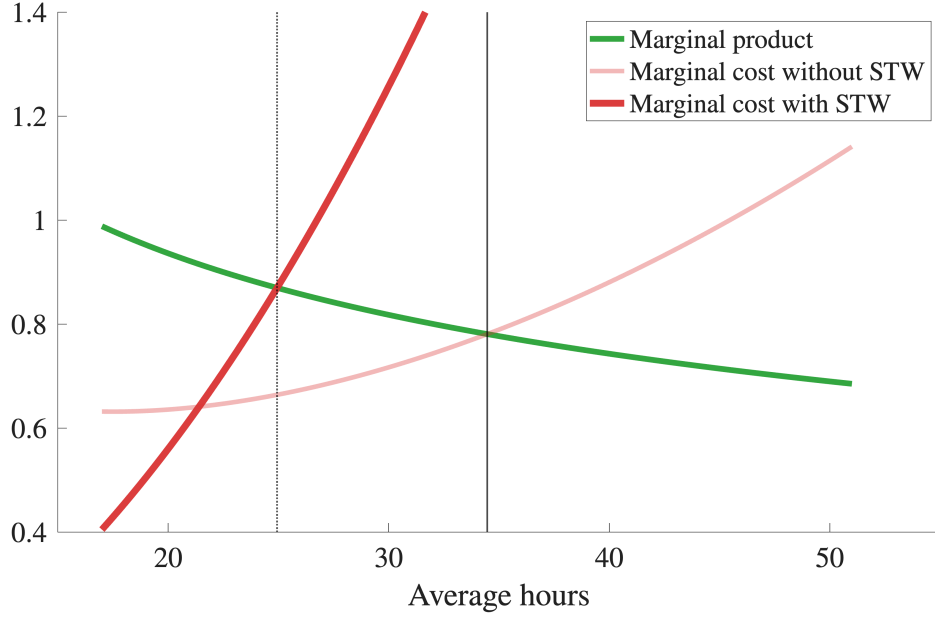


Figure 4: Choice of hours

Note.— The marginal product is the additional output generated by an increase in hours per worker. Marginal cost is the additional compensation that the firm must pay each worker when average hours increase: $\omega_h(s, h; S)$ without short-time work and $\tilde{\omega}_h(s, \tilde{h}; S)$ under the policy. All series are normalized by average hourly compensation at the optimal no-policy hours. The solid and dotted vertical lines indicate optimal hours without and with short-time work, respectively. The aggregate state S was kept fixed and we abstract from eligibility and participation choices for this figure.

worked decrease only slightly because large deviations from ‘baseline hours’ are costly, as discussed in Section 3.2. Firms therefore adjust labor demand primarily along the extensive margin, i.e., by reducing the number of workers, thereby increasing unemployment. The model generates a persistent increase in unemployment of about half a percentage point. The hump-shaped response is the result of the presence of search frictions and adjustment costs.

A key mechanism that limits the increase in unemployment is the endogenous increase in the vacancy-filling rate q . As unemployment rises relative to vacancies, labor-market tightness falls. The resulting increase in q lowers the expected cost c_v/q of hiring a worker and induces expanding firms to post more vacancies. These expanding firms are those with relatively high current idiosyncratic productivity draws. Workers are therefore reallocated toward more productive firms, which improves allocative efficiency, as shown by the fall in the standard deviation of firms’ marginal products of labor in the bottom row of Figure 5.

We now contrast this with the case when short-time work is available, shown as the red dashed lines in Figure 5. The intended effect of the policy is to make reductions in hours less costly. The fall in labor demand is now absorbed by reductions in average hours worked, which fall sharply, while firms retain workers they would otherwise dismiss. In addition, there is an option value of using short-time work in future peri-

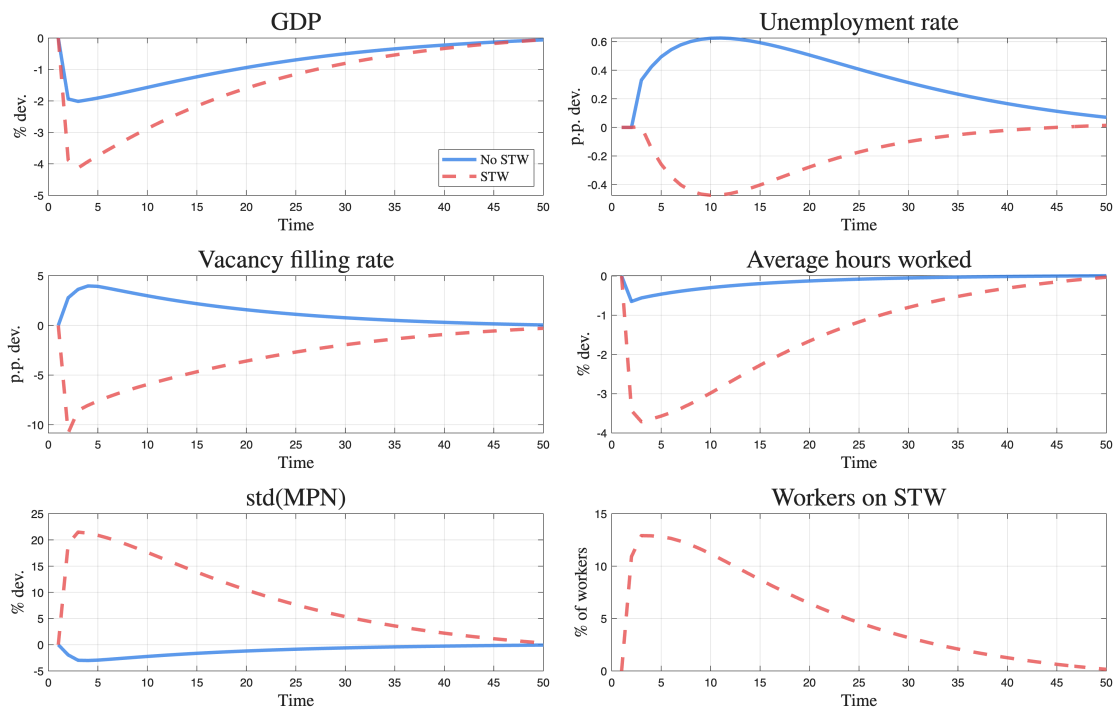


Figure 5: Impulse response functions

Note.— The solid blue (dashed red) lines show a recession without (with) short-time work. In both cases, aggregate productivity A falls by 1.5%. $std(MPN)$ refers to the cross-sectional standard deviation in marginal products of labor. The plotted share of workers on STW is an unconditional impulse response and peaks at about 12.5%. The corresponding first-year average conditional on the recession and the policy remaining in place is 15.6% in the model, close to its empirical calibration target of 15.4%.

ods, which provides an additional incentive to retain existing workers. Taken together, unemployment falls, reaching about 0.45 percentage point below its initial level after roughly ten months and about 0.9 percentage point below the no-STW level after one year.

The policy has an important general equilibrium consequence. With fewer workers entering unemployment, labor-market tightness rises and the vacancy-filling rate q falls. When a firm decides not to fire a worker, the firm inadvertently decreases the vacancy-filling probability for all other firms. An individual firm does not take this externality into account. In equilibrium, the cost c_v/q of hiring a worker increases, leading to fewer vacancies and a reduction in allocative efficiency. We illustrate this increase in factor misallocation through the cross-sectional standard deviation of marginal products of labor, shown in the bottom-left panel. While this measure of misallocation falls during recessions without short-time work, it increases by about 20% with the policy, leading to lower measured aggregate productivity. The sharp reduction in hours and the deterioration in allocative efficiency exacerbate the fall in output from about 2.0% without short-time work to 4.1% with the policy.

Taken together, the results in Figure 5 show that short-time work stabilizes employ-

ment during a recession, but reduces allocative efficiency and aggregate output.

3.4. The role of the endogenous vacancy filling rate

To highlight the importance of the endogenous response of the vacancy filling rate q in general equilibrium, we compute impulse responses for an alternative, partial equilibrium economy in which q remains exogenously fixed at its steady-state value. The remainder of the calibration is identical to the benchmark model discussed above.

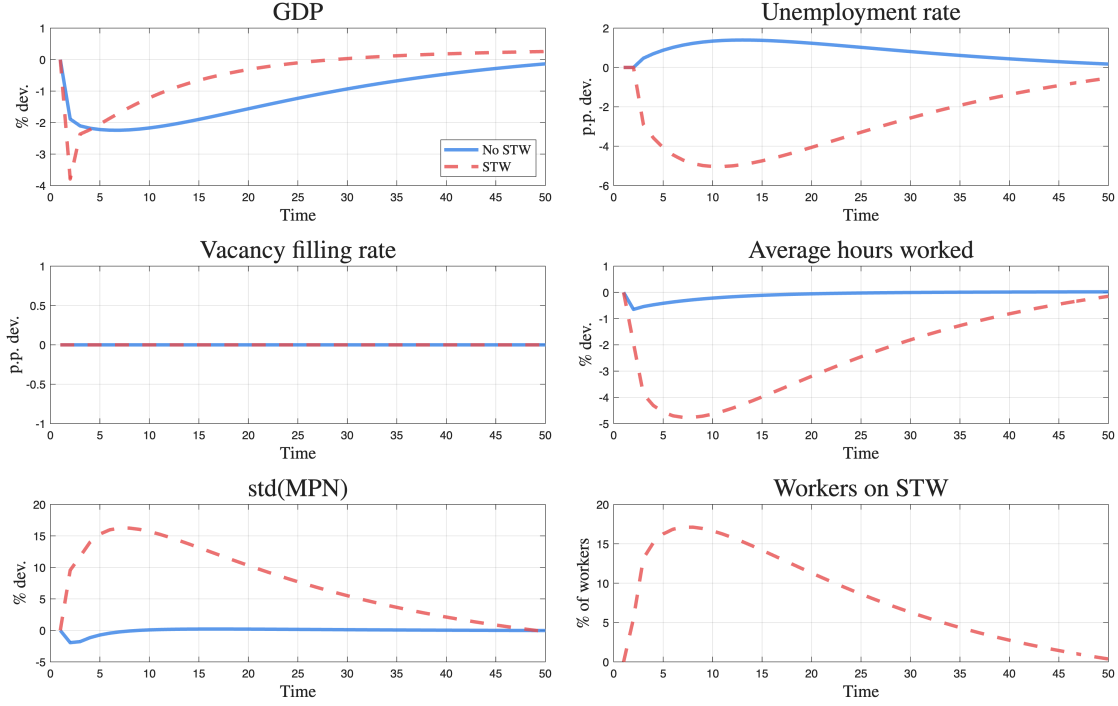


Figure 6: Impulse response functions in partial equilibrium

Note.— The solid blue (dashed red) lines show a recession without (with) short-time work. In both cases, aggregate productivity A falls by 1.5%. $std(MPN)$ refers to the cross-sectional standard deviation in marginal products of labor. The vacancy filling rate q is exogenously held fixed at its steady-state value.

The results, shown in Figure 6, highlight several important differences compared to the general equilibrium exercise above. Absent the policy (blue, solid lines), the increase in unemployment is now *stronger*, with the unemployment rate increasing by about an additional percentage point. With an exogenous vacancy-filling rate, hiring no longer becomes cheaper during recessions. This makes it relatively more costly for firms to hire workers, which lowers job creation and increases the unemployment rate, while also resulting in a slightly larger drop in GDP.

The dashed red lines show the partial equilibrium response to a recession with short-time work. The large decrease in the vacancy-filling rate that was visible in Figure 5 is now gone. This makes hiring relatively less costly in partial equilibrium, resulting in

more job creation and an even stronger decrease in the unemployment rate, leading to a milder and shorter decline in GDP. The hours response is roughly similar in general and partial equilibrium because it is largely determined by the unchanged generosity parameter δ . Our measure of factor misallocation, the standard deviation in marginal products of labor across firms, shows only about half of the increase observed in general equilibrium. This is a direct consequence of the fact that with a constant q , labor can continue to flow to hiring firms without an increase in costs.

4. Conclusion

This paper studies the employment and productivity effects of short-time work in Germany. The policy facilitates reductions in average hours worked with the goal of preventing layoffs. Using German micro-level data, we estimate a search model with heterogeneous multi-worker firms. We find that *Kurzarbeit* leaves unemployment about 0.9 percentage point below the no-policy path after one year, but deepens the output trough from about 2.0% to 4.1%. This additional decline reflects both the policy-induced reduction in hours and a deterioration in allocative efficiency. The latter is amplified in general equilibrium because worker retention reduces the vacancy-filling rate and makes expansion more costly for productive firms.

Our analysis abstracts from job-to-job transitions and worker heterogeneity. Allowing employed workers to move directly between firms could affect the strength and speed of the reallocation channel, while worker heterogeneity would allow the policy’s employment and distributional effects to vary with worker characteristics and match quality. Incorporating these margins constitutes a fruitful avenue for future research.

References

- ACEMOGLU, D. AND W. B. HAWKINS (2014): “Search with multi-worker firms,” *Theoretical Economics*, 9, 583–628.
- BACHMANN, R. AND C. BAYER (2014): “Investment Dispersion and the Business Cycle,” *American Economic Review*, 104, 1392–1416.
- BALLEER, A., B. GEHRKE, W. LECHTHALER, AND C. MERKL (2016): “Does short-time work save jobs? A business cycle analysis,” *European Economic Review*, 84, 99–122.
- BARTELSMAN, E., J. HALTIWANGER, AND S. SCARPETTA (2013): “Cross-Country Differences in Productivity: The Role of Allocation and Selection,” *American Economic Review*, 103, 305–334.
- BAUER, A. AND I. KING (2018): “The Hartz reforms, the German Miracle, and labor reallocation,” *European Economic Review*, 103, 1–17.

- BOERI, T., H. BRUECKER, N. FUCHS-SCHÜNDEIN, AND T. MAYER (2011): “Short-time work benefits revisited: some lessons from the Great Recession,” *Economic Policy*, 26, 697–765.
- BRENKE, K., U. RINNE, AND K. ZIMMERMANN (2013): “Short-time work: The German answer to the Great Recession,” *International Labour Review*, 152, 287–305.
- BURDA, M. C. AND J. HUNT (2011): “What Explains the German Labor Market Miracle in the Great Recession,” *Brookings Papers on Economic Activity*, 42, 273–335.
- BURDETT, K. AND R. WRIGHT (1989): “Unemployment Insurance and Short-Time Compensation: The Effects on Layoffs, Hours per Worker, and Wages,” *Journal of Political Economy*, 97, 1479–1496.
- CAHUC, P. AND S. CARCILLO (2011): “Is Short-Time Work a Good Method to Keep Unemployment Down?” *Nordic Economic Policy Review*, 1, 133–169.
- CAHUC, P., F. KRAMARZ, AND S. NEVOUX (2018): “When Short-Time Work Works,” Report.
- COOPER, R., J. HALTIWANGER, AND J. L. WILLIS (2007): “Search frictions: Matching aggregate and establishment observations,” *Journal of Monetary Economics*, 54, 56–78.
- COOPER, R. W. AND I. SCHOTT (2023): “Capital Reallocation and the Cyclicity of Aggregate Productivity,” *Quantitative Economics*, 14, 1337–1365.
- COSAR, A. K., N. GUNER, AND J. TYBOUT (2016): “Firm Dynamics, Job Turnover, and Wage Distributions in an Open Economy,” *American Economic Review*, 106, 625–663.
- DEN HAAN, W. J. (2010): “Assessing the accuracy of the aggregate law of motion in models with heterogeneous agents,” *Journal of Economic Dynamics and Control*, 34, 79–99.
- DUSTMANN, C., B. FITZENBERGER, U. SCHÖNBERG, AND A. SPITZ-OENER (2014): “From Sick Man of Europe to Economic Superstar: Germany’s Resurgent Economy,” *Journal of Economic Perspectives*, 28, 167–188.
- ELSBY, M. W. L. AND R. MICHAELS (2013): “Marginal Jobs, Heterogeneous Firms, and Unemployment Flows,” *American Economic Journal: Macroeconomics*, 5, 1–48.
- FUJITA, S. AND M. NAKAJIMA (2016): “Worker Flows and Job Flows: A Quantitative Investigation,” *Review of Economic Dynamics*, 22, 1–20.
- GEHRKE, B. AND B. HOCHMUTH (2020): “Counteracting Unemployment in Crises: Non-Linear Effects of Short-Time Work Policy,” *The Scandinavian Journal of Economics*, n/a.

- GIUPPONI, G. AND C. LANDAIS (2022): “Subsidizing Labour Hoarding in Recessions: The Employment and Welfare Effects of Short-time Work,” *The Review of Economic Studies*, 90, 1963–2005.
- HARTUNG, B., P. JUNG, AND M. KUHN (2026): “Unemployment Insurance Reforms and Labour Market Dynamics,” *The Review of Economic Studies*, 93, 517–555.
- HOFFMANN, F. AND T. LEMIEUX (2016): “Unemployment in the Great Recession: A Comparison of Germany, Canada, and the United States,” *Journal of Labor Economics*, 34, S95–S139.
- HSIEH, C.-T. AND P. J. KLENOW (2009): “Misallocation and Manufacturing TFP in China and India,” *The Quarterly Journal of Economics*, 124, 1403–1448.
- JUNG, P. AND M. KUHN (2014): “Labour Market Institutions and Worker Flows: Comparing Germany and the US,” *The Economic Journal*, 124, 1317–1342.
- KOHLBRECHER, B., C. MERKL, AND D. NORDMEIER (2016): “Revisiting the matching function,” *Journal of Economic Dynamics and Control*, 69, 350–374.
- KOPP, D. AND M. SIEGENTHALER (2021): “Short-Time Work and Unemployment in and after the Great Recession,” *Journal of the European Economic Association*, 19, 2283–2321.
- KRAUSE, M. U. AND H. UHLIG (2012): “Transitions in the German labor market: Structure and crisis,” *Journal of Monetary Economics*, 59, 64–79.
- KRUSELL, P. AND J. A. SMITH (1998): “Income and Wealth Heterogeneity in the Macroeconomy,” *Journal of Political Economy*, 106, 867–896.
- KUDOH, N., H. MIYAMOTO, AND M. SASAKI (2019): “Employment and hours over the business cycle in a model with search frictions,” *Review of Economic Dynamics*, 31, 436–461.
- KUDOH, N. AND M. SASAKI (2011): “Employment and hours of work,” *European Economic Review*, 55, 176–192.
- ROYS, N. (2016): “Persistence of shocks and the reallocation of labor,” *Review of Economic Dynamics*, 22, 109–130.
- SMITH, E. (1999): “Search, Concave Production, and Optimal Firm Size,” *Review of Economic Dynamics*, 2, 456–471.
- STOLE, L. A. AND J. ZWIEBEL (1996): “Intra-Firm Bargaining under Non-Binding Contracts,” *The Review of Economic Studies*, 63, 375–410.

A. Institutional details on short-time work in Germany

Short-time work is a job-retention program that allows firms facing a temporary reduction in labor demand to reduce hours per worker while preserving existing employment relationships. Participating firms pay workers for the hours actually worked, while the government replaces part of the associated earnings loss. The regular replacement rate mirrors unemployment insurance: 60% of lost net earnings for workers without children and 67% for workers with children. During the Covid recession, replacement rates were increased successively with the duration of participation. Although short-time work is also available in response to normal cyclical fluctuations, its primary purpose is to prevent layoffs during severe downturns.

The German Code of Social Law, Book 3 (SGB III), distinguishes short-time work due to economic distress (§170), seasonal short-time work (§175), and transfer short-time work, used primarily in restructuring episodes (§216b). The short-time work expansion studied in this paper concerns short-time work due to economic distress. If an application is approved, a firm can reduce employees' working hours and compensation while affected workers receive government benefits. Without such approval, firms remain subject to contractual and collectively agreed limits on reductions in hours.

Approval is subject to several requirements. Before the 2009 expansion, the reduction in hours had to be "unavoidable": firms first had to use accumulated overtime, working-time-account balances, and accrued holidays, and could not use the program if the reduction could be absorbed by working-time flexibility already permitted under employment contracts. At least one-third of a firm's workforce had to suffer an earnings loss of at least 10%. Firms also had to demonstrate that the reduction in working time was temporary and that full-time employment could be restored after the program ended; the regular maximum duration was six months.

In response to the 2009 recession, the government loosened the workforce and temporariness requirements, extended the maximum duration of the program, and broadened access to part-time workers and apprentices. Initially, participating firms continued to pay employers' social-security contributions in full. Subsequent modifications made these contributions proportional to hours actually worked and, in some cases, shifted them entirely to the government. These changes lowered both the administrative and financial cost of participation and contributed to the exceptional use of short-time work during the recession.

Several papers have studied features of German labor-market institutions that help explain the country's performance during the 2009 and Covid recessions; see Burda and Hunt (2011), Brenke, Rinne, and Zimmermann (2013), and Dustmann et al. (2014) for broader institutional accounts. Burda and Hunt (2011) emphasize restrained hiring during the preceding expansion, wage moderation, and flexibility in hours through short-time work and working-time accounts. Working-time accounts generally have to be balanced within a specified period, and firms had to exhaust positive balances and accrued holidays before becoming eligible for short-time work. Burda and Hunt (2011) show that short-time work was the single most important source of changes in hours per worker in 2008 and 2009 (*cf.* their Table 7). Dustmann et al. (2014) additionally

emphasize the decline in union coverage and collective wage agreements beginning in the 1990s. The *Hartz* reforms implemented in the mid-2000s reduced the generosity of long-term unemployment insurance, but did not directly change employers’ flexibility to adjust average hours (Burda and Hunt, 2011; Krause and Uhlig, 2012; Hoffmann and Lemieux, 2016; Bauer and King, 2018; Hartung, Jung, and Kuhn, 2026).

B. Data Appendix

This appendix describes the German micro data and presents additional empirical results.

B.1. Plant-level data

Our micro data come from the ‘AFiD-Panel Industriebetriebe’, a confidential panel of the universe of manufacturing plants based in Germany and administered by the German Federal Statistical Office in cooperation with the statistical offices of the German *Länder*.¹⁵ Participation in the monthly surveys is mandatory. The panel contains annual information from plants with more than 20 employees between 1995 and 2010. The data include smaller plants if they belong to a *firm* with at least 20 employees.¹⁶ The data covers up to 68,000 plants per year. We use an unbalanced panel of 39,180 plants. We start with 59,268 unique plants before winsorizing the data. Plants with fewer than three consecutive observations account for 2.8 percent of all plant-year observations and were deleted from the final sample. This leaves an average of 39,180 plants per year. Of those, 19,373 plants were observed in every year between 1995 and 2010.

Our focus on the manufacturing sector is warranted for the purpose of this paper because the use of STW, especially in 2009, was heavily concentrated in that sector. At the height of the STW program in 2009, over 80% of workers and 45% of plants that used STW were operating in the manufacturing sector and 16.9% of all manufacturing employees were on *Kurzarbeit*. In Germany, manufacturing employment is heavily concentrated in medium-sized and/or family-owned firms (German *Mittelstand*). Including these firms is therefore crucial in any assessment of labor market policies.

Descriptive statistics are shown in Table B.1. The table shows time-series averages of total employment, total hours, hours per worker, and revenues. The average number of employees is 98.5. More than 50% of plants employ less than 50 workers, whereas the largest 10% employ more than 220 workers. Average hours worked at the plant level are computed as total hours worked divided by total employment. The large differences between the 10th and the 90th percentile of the distribution of revenues illustrate substantial plant heterogeneity. The 2009 recession is clearly visible in our micro data: Between 2008 and 2009 mean (median) revenues fell by 9% (12%).

¹⁵A detailed description of the data can be found at DOI:10.21242/42111.2021.00.01.1.1.0.

¹⁶From 2007 to 2010, the cutoff for inclusion in the survey was increased to 50 employees. In our data, this change did not significantly influence the distribution of our variables of interest, partly because many smaller plants continued to provide information.

	Count	Mean	SD	IQR	p10	Median	p90
<i>N</i>	38,839	98.5	142.6	73.8	19.4	48.2	228.0
<i>H</i>	33,617	156,300	20,576	11,694	3,578	8,366	35,107
<i>H/N</i>	34,303	135.8	35.7	31.6	104.5	134.0	167.9
<i>PY</i>	39,180	1,531,785	3,106,538	1,116,285	101,242	474,343	3,766,944

Table B.1: Summary Statistics

Note.— Summary statistics for Employment *N*, Hours *H*, Hours per Employee *H/N*, and Revenue *PY*. The table shows average values over all years. Revenues are deflated to 2005 Euros. *Source: AFid Panel Industriebetriebe* and authors' calculations.

The moments we use to calibrate the model are almost identical in the balanced and unbalanced panels of our micro data due to the fact that entry and exit rates are very low. We find no evidence for a significant change in entry and exit rates in our data during the years of the STW policy extension. For this reason, the model abstracts from firm entry and exit.

B.2. Additional empirical results

Figure B.1 shows the same three time series as Figure 1, but focuses on the time period around the Covid pandemic. A similar pattern emerges: While GDP growth was more negatively affected in Germany, the U.S. unemployment rate increased considerably more. At the same time, labor productivity declined markedly more in Germany. One key difference between 2020 and 2009 is that the German unemployment rate prior to Covid was only 3.5%, whereas it was over 9% in 2009. The re-emergence of similar unemployment and productivity dynamics in 2020, when the transition dynamics that may have influenced the German labor market in 2009 had largely run their course, suggests that the use of *Kurzarbeit* contributed to this pattern.

In Figures 1 and B.1 we compared labor productivity in Germany and the U.S. Figure B.2 shows that these patterns are even more pronounced in the manufacturing sector, which was the German sector where *Kurzarbeit* was most heavily used, particularly during the 2009 recession.

C. Model Appendix

In this appendix we derive key model results and show additional computational results.

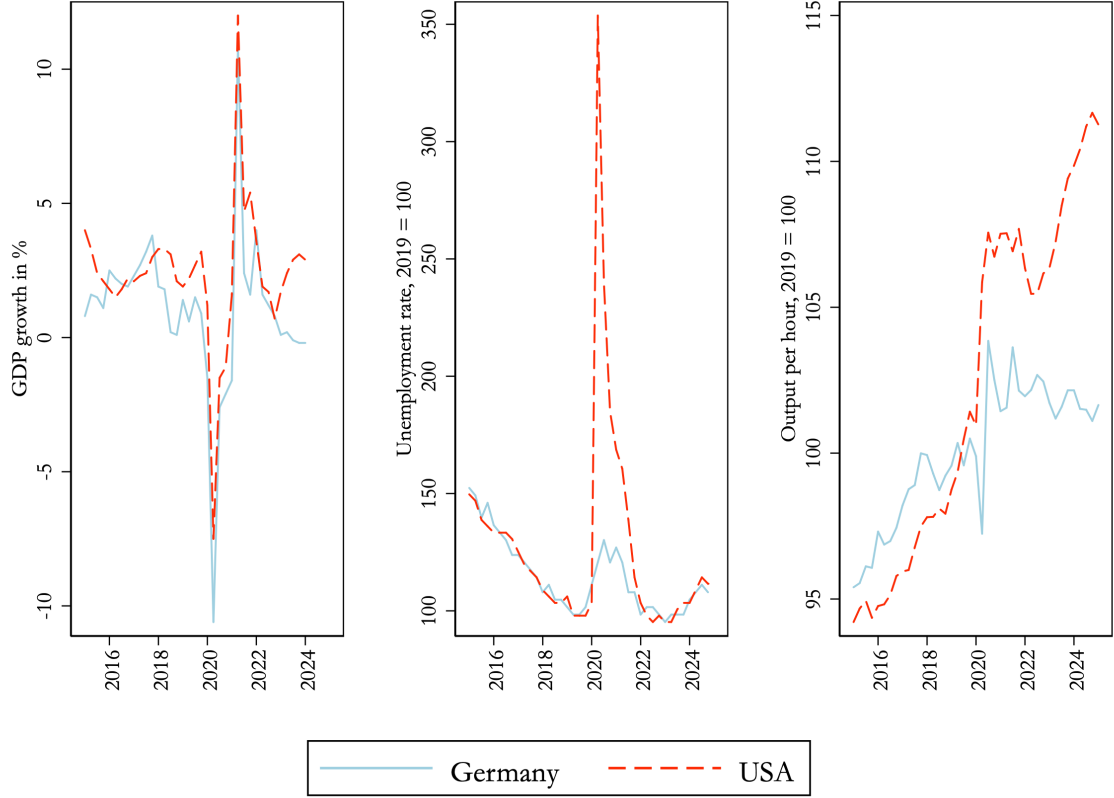


Figure B.1: GDP growth and unemployment rates

Note.— GDP growth rates are in % compared to the same period one year ago. Unemployment rates and labor productivity are normalized to 100 in 2019. *Source:* OECD, BLS, German Statistical Office.

C.1. Wage setting

We now derive the compensation function used in the main text. The firm's optimal employment policy function in (12) can be written as

$$n'(\varepsilon, n; S) = \begin{cases} \bar{n}(\varepsilon, n; S) & \text{if } \varepsilon > \bar{\varepsilon}(n, S) \\ n & \text{if } \underline{\varepsilon}(n, S) < \varepsilon < \bar{\varepsilon}(n, S) \\ \underline{n}(\varepsilon, n; S) & \text{if } \varepsilon < \underline{\varepsilon}(n, S), \end{cases} \quad (\text{C.21})$$

where $\bar{\varepsilon}$ and $\underline{\varepsilon}$ are productivity thresholds. We now show that the compensation function for firms that are hiring, firing, or inactive is identical.

Hiring firms Recall from equation (2) that the value of an unemployed worker at the beginning of the period is given by

$$W^u(S) = b + \beta \mathbb{E}_{S'|S} \left[(1 - \phi) W^u(S') + \phi \mathbb{E}_{s'|S'}^v W^e(s'; S') \right], \quad (\text{C.22})$$

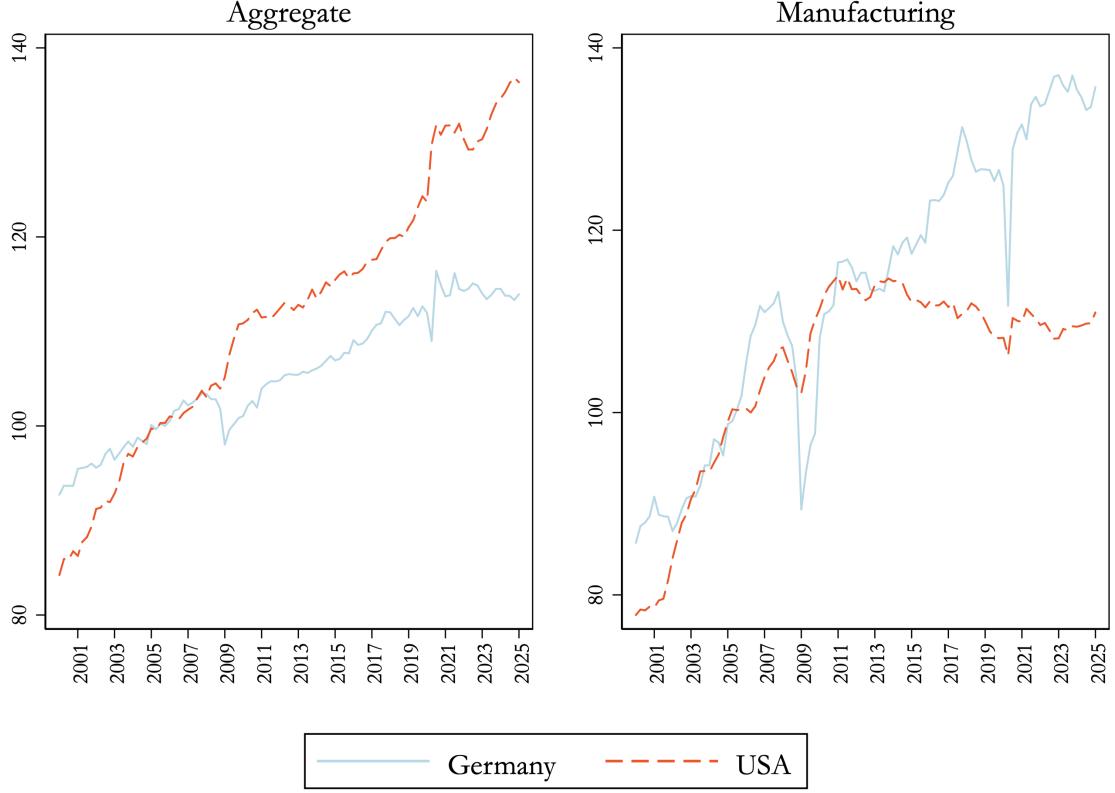


Figure B.2: Labor productivity - Aggregate economy and manufacturing

Note.— Labor productivity is normalized to 100 in 2005. It is defined as output per hour. *Source:* BLS, German Statistical Office.

where $\mathbb{E}^v$ denotes an expectation over the vacancy-weighted distribution of firms that are hiring. Thus, W^u depends only on the aggregate state and not on the state of a particular hiring firm.

Similarly, an employed worker in a firm that is posting additional vacancies will be employed next period and has the value function

$$W^e(\varepsilon, n; S) = \omega(\varepsilon, n, h; S) - e(h) + \beta \mathbb{E}_{\varepsilon', S' | \varepsilon, S} W^e(\varepsilon', \bar{n}(\varepsilon, n; S); S'). \quad (\text{C.23})$$

Under Nash sharing the difference between (C.23) and (C.22) must be equal to a hiring firm's marginal surplus. This surplus is given by (13) in the main text. Using $\phi = q\theta$, the Nash sharing rule in (2.3) can thus be written as follows:

$$\begin{aligned} & \omega(\varepsilon, n, h; S) - e(h) - b + \beta \mathbb{E}_{\varepsilon', S' | \varepsilon, S} [W^e(\varepsilon', \bar{n}(\varepsilon, n; S); S') - W^u(S')] \\ & \quad - q\theta \beta \mathbb{E}_{S' | S} \mathbb{E}_{s' | S'}^v [W^e(s'; S') - W^u(S')] \\ & = \frac{\eta}{1 - \eta} \left[A\varepsilon \alpha h^\alpha n^{\alpha-1} - \omega(\varepsilon, n, h; S) - \omega_n(\varepsilon, n, h; S)n + \frac{c_v}{q} \right], \end{aligned} \quad (\text{C.24})$$

For the current hiring firm, applying Nash sharing to the continuation surplus and using (6) gives $\frac{\eta}{1-\eta} \frac{c_v}{q}$. The same relation holds for every hiring firm and hence for the vacancy-weighted expectation in the second continuation term. Substituting these expressions into (C.24) gives the compensation function

$$\omega(\varepsilon, n, h; S) = (1 - \eta)(e(h) + b) + \eta [A\varepsilon\alpha h^\alpha n^{\alpha-1} - \omega_n(\varepsilon, n, h; S)n] + \eta c_v \theta. \quad (\text{C.25})$$

Firing firms The value of a worker in a firing firm is

$$\begin{aligned} W^e(\varepsilon, n; S) &= \omega(\varepsilon, n, h; S) - e(h) \\ &\quad + \beta \mathbb{E}_{\varepsilon', S' | \varepsilon, S} [(1 - s)W^e(\varepsilon', \underline{n}(\varepsilon, n; S); S') + sW^u(S')]. \end{aligned} \quad (\text{C.26})$$

There is a probability that the worker will be unemployed next period. With the counter-probability, the worker will remain employed in the firm, which will continue operating with $\underline{n}(\varepsilon, n; S)$ employees. A firm's threat point consists of losing the marginal worker. Using the same steps as above, we obtain

$$\begin{aligned} \omega(\varepsilon, n, h; S) - e(h) - b &+ (1 - s)\beta \mathbb{E}_{\varepsilon', S' | \varepsilon, S} [W^e(\varepsilon', \underline{n}(\varepsilon, n; S); S') - W^u(S')] \\ &- q\theta\beta \mathbb{E}_{S' | S} \mathbb{E}_{s' | S'}^v [W^e(s'; S') - W^u(S')] \\ &= \frac{\eta}{1 - \eta} [A\varepsilon\alpha h^\alpha n^{\alpha-1} - \omega(\varepsilon, n, h; S) - \omega_n(\varepsilon, n, h; S)n]. \end{aligned} \quad (\text{C.27})$$

Rearranging this equation yields

$$\omega(\varepsilon, n, h; S) = (1 - \eta)(e(h) + b) + \eta [A\varepsilon\alpha h^\alpha n^{\alpha-1} - \omega_n(\varepsilon, n, h; S)n] + \eta \theta c_v. \quad (\text{C.28})$$

Note that this compensation function takes the same form as the compensation function for hiring firms in (C.25).

Inactive firms Finally, consider the case of a worker in an inactive firm.

$$W^e(\varepsilon, n; S) = \omega(\varepsilon, n, h; S) - e(h) + \beta \mathbb{E}_{\varepsilon', S' | \varepsilon, S} W^e(\varepsilon', n; S'). \quad (\text{C.29})$$

Nash sharing implies

$$\begin{aligned} \omega(\varepsilon, n, h; S) - e(h) - b &+ \beta \mathbb{E}_{\varepsilon', S' | \varepsilon, S} [W^e(\varepsilon', n; S') - W^u(S')] - q\theta\beta \mathbb{E}_{S' | S} \mathbb{E}_{s' | S'}^v [W^e(s'; S') - W^u(S')] \\ &= \frac{\eta}{1 - \eta} [A\varepsilon\alpha h^\alpha n^{\alpha-1} - \omega(\varepsilon, n, h; S) - \omega_n(\varepsilon, n, h; S)n + \mathbb{E}_{\varepsilon', S' | \varepsilon, S} \beta V_n(\varepsilon', n; S')]. \end{aligned} \quad (\text{C.30})$$

Solving for the compensation function yields

$$\omega(\varepsilon, n, h; S) = (1 - \eta)(e(h) + b) + \eta [A\varepsilon\alpha h^\alpha n^{\alpha-1} - \omega_n(\varepsilon, n, h; S)n] + \eta \theta c_v. \quad (\text{C.31})$$

Note that this is again the same compensation function as in (C.25).

C.2. Additional Quantitative Results

Calibration details. We jointly choose $\Theta = (\rho_\varepsilon, \sigma_\varepsilon, \mu_\varepsilon, e_1, b, c_v)$ to match the seven moments in the upper part of Table 2. At every candidate value of b and e_1 , the scale parameter of work disutility is computed from the average-hours normalization $\bar{h} = 34$ as

$$e_0 = \frac{1 + e_1}{e_1} b \bar{h}^{-(1+e_1)}.$$

For each candidate parameter vector, we solve the stationary firm problem and its invariant distribution. We then minimize the unweighted mean absolute proportional deviation between the selected model moments, $m_j(\Theta)$, and their empirical counterparts, $\hat{m}_j$:

$$\mathcal{D}(\Theta) = \frac{1}{7} \sum_{j=1}^7 \left| \frac{m_j(\Theta) - \hat{m}_j}{\hat{m}_j} \right|.$$

The annual changes in hours and employment used to construct the inaction moments are generated from the model's twelve-month transition matrix. In both the model and the data, a small adjustment is an annual change of no more than 2.5% in absolute value.

Although all parameters affect several moments in equilibrium, some mappings are particularly informative. The unconditional mean of log productivity, μ_ε , shifts the scale of firms and is therefore closely tied to the average establishment-size target of 98.5 workers. The innovation volatility σ_ε and persistence ρ_ε jointly govern the dispersion and dynamics of firm productivity and hence the frequency with which firms adjust hours and employment. In particular, greater persistence makes current productivity more informative about future productivity and changes firms' willingness to incur the cost of adjusting employment.

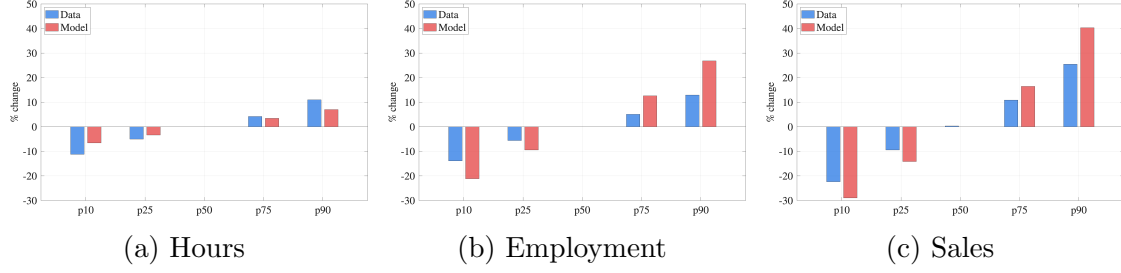
The curvature of work disutility, e_1 , governs the responsiveness of hours and the tradeoff between the intensive and extensive margins. It is therefore informed by average hours, the share of small hours changes, and the ratio $cv(n)/cv(h)$. The vacancy cost c_v makes employment adjustment costly and shifts labor-demand adjustment toward hours; the relative dispersion of employment and hours, the share of small employment changes, and equilibrium worker flows are consequently informative about it. Finally, b affects workers' outside option and match surplus. The replacement-rate moment, defined as b relative to average compensation, is especially informative about b .

The unemployment rate is one of the calibration targets. For each candidate parameter vector, the stationary unemployment stock implied by the firm distribution is also required to be internally consistent with the unemployment inflows and outflows generated by firms' policy functions and the matching technology. At the estimated parameters, the model produces an unemployment rate of 9.85%, compared with 9.04% in the data.

As an untargeted validation exercise, Figure C.1 compares selected quantiles of the annual firm-level changes in hours, employment, and sales. The model reproduces the zero median of the hours and employment change distributions. It understates the

dispersion of hours changes, however, while overstating the dispersion of employment and sales changes. The latter discrepancy illustrates a limitation of the model's parsimonious adjustment-cost structure: conditional on adjusting employment, firms make larger changes than in the data.

Figure C.1: Distributions of firm-level changes in hours, employment, and sales



Note.— The bars report selected quantiles of annual establishment-level changes. The blue bars show the data and the red bars show the model. These quantiles are not targeted in the calibration.

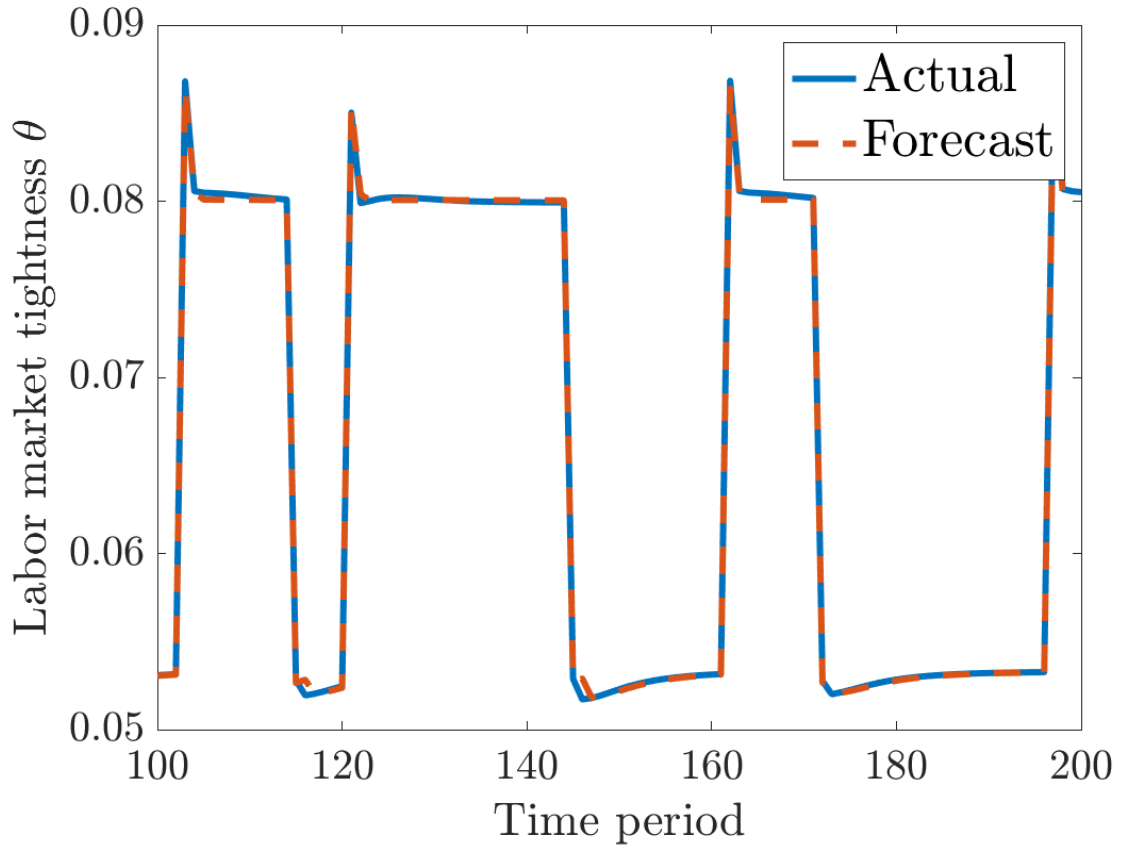


Figure C.3: Assessing forecast accuracy

Figure compares the actual law of motion of θ (blue solid line) to the model-consistent forecast (red dashed line) for a sample of a larger simulation. The R^2 underlying the forecast was in excess of 0.999. The maximum forecast error was 0.007%.